

Prediction of Land Suitability for Crop Cultivation Based on Soil and Environmental Characteristics Using Modified Recursive Feature Elimination Technique with Various Classifiers

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Abstract: Predicting which crops will do well in a given area is an important part of modern farming. Soil qualities, weather conditions (like temperature and rainfall), and nutrient levels (especially nitrogen and phosphorus) can all have an effect. To make accurate predictions, these complicated environmental and soil factors must be looked at, since the same foods can't grow well in all areas. By finding patterns in these parameters, ML methods offer a reliable way to find the best crops. In this situation, FS is very important for picking out the most important traits, which leads to accurate predictions and faster computations. A “Modified Recursive Feature Elimination (MRFE)” method is used to pick out and rank the most important features from the information in a way that makes predictive models work better. Several classifiers are used on the chosen features to find out which crop is best, such “as KNN, Naive Bayes, Decision Tree, Support Vector Machine (SVM), Random Forest, Bagging Classifier, and Voting Classifier. Some of the measures used to judge performance are Accuracy (ACC), Precision, Recall, Specificity, F1-Score, Area Under the Curve (AUC), Mean Absolute Error (MAE)”, and Log Loss. The test results show that MRFE is good at finding important features, and the Voting Classifier does the best of all of them, getting a perfect score of 100%, which shows how stable and reliable it is for accurate crop prediction.

“Index Terms: Agriculture, Radio frequency, Soil, Predictive model, Vegetation, Feature extraction, Support vector machines”.

1. INTRODUCTION

Food security and economic peace are still two of the most important things that agriculture does. As the world's population grows quickly and arable land is used up faster, it has become very important for modern farming methods to find the best ways to grow crops [1]. Farmers' real-world information and experience with the area are often used in traditional farming methods. They are useful, but not sufficient to overcome the problems that are become even more problematic due to climate change, soil degradation, and limited resources [2]. The additional inventions in technology using data have

rendered it possible to allow farmers to make improved judgments through the integration of the data concerning the weather, the environment and the land [3]. Machine learning and predictive analytics have been very useful in this area. They allow people to make sound forecasts about about the success of some crops and the quantity of profit they will produce in different areas and in different climatic conditions [4]. Through such technologies, farmers can probably be in a better position to make decisions about the kind of crops they grow, the quantity of water they apply, and how

to handle fertilizer. This results in high quality crops with increased life span.

Notwithstanding the considerable advancement that is already made, modern agricultural predictive systems still face the difficulties with the administration of massive amounts of data and the identification of the most relevant features to be applied to the intended goal of the development of accurate predictions regarding crop suitability [5]. A large number of existing techniques cannot work in every kind of land and climatic conditions and do not take into account the connection between items which leads to poor results [6]. Also, one cannot be sure of crops growing in particular regions because of the change in soil moisture, PH level, rainfall, and nutrient concentration [7]. The problems indicate that there is a need to have more sophisticated means of analysis that are capable of detecting the most influential factors that determine results and at the same time handle a substantial volume of information. These problems should be eliminated to come up with prediction models that are both general and localized. These models can then help the government officials and producers to make informed decisions about agriculture.

The main aim of the study is to design an intelligent and efficient mechanism of classifying the land in terms of its suitability to support agriculture by studying the natural characteristics and the soil structures of the land. The main aim is to have systematic methodology which can determine, explore and apply decisive factors that affect crop growth without having to resort to extra or irrelevant data [8]. It is hoped that the proposed method is able to increase the precision of the predictions by carefully choosing the most important features and using them in a systematic way of modeling. The current effort is trying to create a reliable system that can be adapted to the changing nature of farm

conditions and is also accurate and easy to use through thorough analysis and study of its performance.

This integration is necessary since it can transform the crop planning process by making it more flexible and dependent on more information [9]. This strategy can contribute to the effectiveness of land use by farmers, a decrease in resource usage, and the minimization of risks attributed to uncertainty in weather patterns through the increase in efficiency in crop fit forecasting [10]. It also enables smart farming systems to utilize the real-time data and expand to ensure the health of farms and food security in the long run. This is a big step toward precision agriculture, which is when technology and data science work together to make farming more efficient and better for the environment.

2. LITERATURE REVIEW

“Feature selection (FS)” is an important part of making good predictive models because it lowers the number of dimensions in the data and makes the models easier to understand by picking out the most important features for tasks like classification and regression. Hall and Holmes [11] did one of the first benchmarking studies to compare attribute selection methods for discrete class data mining. They stressed how important it is to find a balance between how fast the computer works and how well the classification works. Their work laid the groundwork for testing FS methods, but it mostly dealt with discrete datasets, which meant it couldn't be used with complex, continuous farming data. Liu and Yu [12] took this idea a step further by suggesting a way to combine FS algorithms for jobs like sorting and grouping. Their results showed that different FS methods work well together. However, their method had trouble scaling up when used on

big, diverse datasets like those found in the agricultural and environmental fields.

Granitto et al. [13] improved variable selection in agro-industrial analysis by using “Recursive Feature Elimination (RFE) along with Random Forest (RF)”. They did this by building on existing FS frameworks. Their results showed that using both RFE and ensemble methods together makes it easier to find important predictive traits. Still, their method was limited to a single field and didn't fully solve the problem of feature association in large datasets. In the same way, Araújo-Azofra and Benítez [14] tested a number of FS techniques on real-world classification problems and found that their results varied a lot based on the characteristics of the data. Even though their work gave us comparative insights, it lacked methodological innovations to deal with correlated or redundant features in a useful way, which can lower the accuracy of predictions in farming settings.

To make it easier to deal with bias in FS, Altmann et al. [15] improved the way they estimated how important features were by adding a permutation-based adjustment method. This method gave more accurate importance scores, but it took a long time to compute for big datasets. Later, Kursa and Rudnicki [16] came up with the Boruta algorithm, which is a strong FS method that uses RF to find all the important features. Boruta was better at being robust and easy to understand, but it often had extra factors that weren't needed, which made the model harder to understand and took longer to process. Ruß and Kruse [17] used FS to predict wheat growth in agriculture, showing how important it is to choose the right soil and climate characteristics. However, their model wasn't able to generalize well because it didn't take into account enough area variation and changing environmental factors.

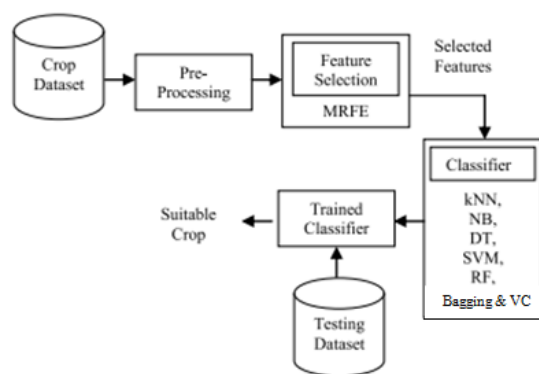
In order to account for correlated factors in genetic data with numerous dimensions, Darst et al. [18] investigated the use of RFE in RF. The association problem was effectively investigated in their study; however, it was not modified to accommodate real-time or geographically distinct datasets. To develop a predictive model of rice blast disease, which is a disease related to agriculture, Hsieh et al. [19] adopted ML and NN. The reason was the inefficiency of the calculations because, in spite of the fact that their framework was rather accurate, a solid FS approach was absent. Finally, Liu et al. [20] utilized both genomic and proteomic data to compare different FS and classification techniques, one after another. They have stressed the need to adopt hybridized schemes that will be effective in handling both duplication and correlation of large and complex datasets.

In spite of these changes, the existing FS approaches are still subject to the limitations, such as the inability to switch to other datasets, expensive nature of operation, and maintenance of redundant information. These difficulties highlight the relevance of an optimized FS structure which can be utilized in weather prediction. This study fills these gaps, by presenting a better feature selection method, which can find out important soil and environmental variables fast, whilst at the same time keeping the model simple, and increasing its predictive capability. This makes the prediction of the success of which commodities to succeed easier.

3. MATERIALS AND METHODS

The suggested system is based on the latest ML technologies to determine the best crops to be grown and evaluate the soil and environmental parameters. The system makes use of numerous data such as the temperature, rainfall, and the amount of nitrogen and phosphorus contained in the fertilizer, not

mentioning the pH, texture, and nutrient level of the soil. MRFE is also used to carefully rank and pick the most important features thus increasing the accuracy and dropping the superfluous features. Subsequently, different classifiers are applied to ascertain the commodities that will perform well in any area. Some of them are “K-Nearest Neighbors (KNN), Naive Bayes, Decision Tree, Support Vector Machine (SVM), Random Forest, Bagging Classifier, and Voting Classifier”. The system guarantees accuracy of the model as it uses the evaluation measures like: “Accuracy, Precision, Recall, F1-Score, AUC, Mean Absolute Error and Log Loss”. This makes sure that the algorithm is capable of producing the right predictions that are reliable on what crops would flourish.



“Fig.1 Proposed Architecture”

The system's design is comprised of a multitude of modules that communicate with one another and utilize information regarding the soil and the surrounding area to determine whether a particular plot of land is suitable for agricultural cultivation. First, you receive the dataset. Then, you clean up the data to fix any missing or mismatched numbers. The MRFE approach is used to choose the best features and make sure that only the most important ones are maintained. After that, multiple ML models are utilized to work with the features that were picked. “Metrics like accuracy, precision, recall, F1-score,

and AUC are used to measure” performance and find the best model.

a) Dataset Collection:

The crop forecast dataset includes different soil and environmental factors that are needed to figure out if the land is suitable for growing crops. Some of the things that make it up are “nitrogen (N), phosphorus (P), potassium (K), pH, electrical conductivity (EC), sulfur (S), organic carbon (OC), and minerals like zinc (Zn), iron (Fe), copper (Cu), manganese (Mn), and boron (B)”. It also includes the type of texture, the season, the amount of rain, and the usual temperature. The preferred crop type, like rice, based on the soil and weather factors is shown by the target variable Crop.

	N	P	K	pH	EC	OC	S	Zn	Fe	Cu	Mn	B	Texture	Season	Rainfall	Avg.Temp	Crop
0	138	8.6	560	7.46	0.62	0.70	5.9	0.24	0.31	0.77	8.71	0.11	Sandy soil	Autumn	202.935536	20.879744	rice
1	213	7.5	338	7.62	0.75	1.06	26.4	0.30	0.86	1.54	2.89	2.29	Sandy soil	Summer	226.655537	21.770462	rice
2	163	9.6	718	7.59	0.51	1.11	14.3	0.30	0.86	1.57	2.70	2.03	Sandy soil	Summer	263.964248	23.004459	rice
3	157	6.8	475	7.64	0.58	0.94	26.0	0.34	0.54	1.53	2.65	1.82	Sandy soil	Autumn	242.864034	26.491096	rice
4	270	9.9	444	7.63	0.40	0.86	11.8	0.25	0.76	1.69	2.43	2.26	Sandy soil	Summer	262.717340	20.130175	rice

Fig.2 CSV Dataset

b) Pre-Processing:

The steps of preprocessing involve planning (investigating, cleaning) of soil and environmental data, feature extraction, and selection. This will make sure that the inputs are of an excellent quality and can be applied to produce accurate, reliable, and fast predictions on the most efficient crops.

Data Processing: Processing data means turning unstructured datasets about crops and soil into a shape that computers can understand. In this step, you get rid of information that isn't needed or is repeated, fix any errors, and organize the data into a single framework. When you handle data properly not only do you maintain the quality and integrity of the inputs but you also reduce noise as well as accelerate the process of making calculations. The standardization of the dataset enables the

opportunity to perform the appropriate analysis and model training in the future. This ensures more accurate predictions are made and ensures that only the most important qualities are considered to determine whether a crop is appropriate.

Data Visualization: Representing data on soil, temperature and crops in form of graphs is referred to as data visualization. It assists individuals in recognizing the tendencies and acquiring knowledge. Visual analysis can make you visualize the relationships, outliers, and trends of data of a great number of various sources. This assists to select the optimum features and enhance the functionality of the model. This step simplifies the process of perceiving and making judgments because it provides simple outlines of complicated data. This assists the researchers to see the bigger picture and ensures that important points that influence crop suitability are appropriately put into consideration during subsequent modeling procedures.

Label Encoding: Under label encoding, you are able to transform categorical objects i.e. the kind of soil or crop into a numerical format which can be comprehended by the ML algorithms. This change is necessary since majority of prediction models take only numbers. This step ensures that the algorithms can process categorical attributes without any bias or misunderstanding of them through enabling a precise mapping of categories to numbers. This initial preprocessing makes the models more cooperative, preserve data on important categories, and generate precise predictions concerning the amount of land to be available given various environmental and soil conditions.

Feature Extraction: To improve the accuracy of ML models The feature extraction is the procedure that gathers significant data when dealing with

massive amounts of dirt and the environment. This step serves to diminish the dimensions of the data though the essential cues that dictate the crop growth quality are retained. It achieves it by finding and creating features that capture the most valuable information. Proper feature extraction eliminates all the additional information, brings to fore the important variables and lets the classifiers concentrate on the major things that drive agricultural output. This makes the model better at forecasting where to plant crops because it is more accurate, faster, and easier to grasp.

Feature Selection Using MRFE: When you use the MRFE approach to choose features, you look at how well they aid with prediction and rank them. This method eliminates rid of the traits that are unneeded or redundant. This simplifies the computation and prevents overfitting. MRFE also produces more accurate models, more efficient models, and classifiers can only make use of the most important data to determine which crops will do the best by paying attention to the most significant soil and weather factors. This step is quite critical in formulating forecasting models which are robust, reliable and simple to comprehend.

c) Training and Testing:

The data that has been pre-processed is divided into two categories in the train-test split. The model is tested and trained in these groups. This phase makes sure that prediction models are only tested on data they haven't seen previously. This gives a clear idea of how well they work in general. Separating training and testing data keeps information from leaking out, makes the model more reliable, and allows you see how well it's doing in terms of accuracy, precision, and other performance indicators. This step is highly critical for creating strong classifiers that can predict what kind of land

will operate best in a lot of different environmental circumstances.

d) Algorithms:

“K-Nearest Neighbors (KNN): KNN is a non-parametric algorithm that can be used for both regression and classification”. By using patterns in local data to make predictions, it makes the system work better. It works by looking for the points that are closest to a certain instance and then giving it a class based on the most common values or the average of those values. When data patterns are well-structured, KNN makes the results more accurate and easier to understand. But how well it works relies on the distance metric and which neighbors are chosen, showing how important it is to have relevant and well-scaled features.

$$distance(x, X_i) = \sqrt{\sum_{j=1}^d (x_j - X_{ij})^2} \quad (1)$$

Naive Bayes: Using Bayes' theorem, NB is a probabilistic classification method that determines the likelihood of each class depending on the features presented. Its job is to make predictions that are accurate across big feature spaces while also being easy on computers. It makes calculations easier and speeds up decision-making by believing that features are conditionally independent. NB often does a good job of generalization, even though it is oversimplified. This is especially true when probabilistic reasoning and feature independence are good estimates of how the data is actually distributed.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \quad (2)$$

Decision Tree: DT are flexible algorithms that divide data back and forth Based on the feature values, a tree-like structure is created where each

leaf node represents a projected class or value. They help models be easier to understand and work better by finding complicated relationships and nonlinear patterns in data. Decision trees make it easy to see how decisions are made and show how important different factors are. Even though they can be overfitted, this problem can be fixed by using trimming or ensemble methods. This makes them more reliable and useful across a wider range of datasets.

$$I(i) = 1 - \sum_{i=1}^k p_i^2 \quad (3)$$

Support Vector Machine (SVM): Help SVM are supervised ML algorithms that find the best hyperplanes that separate classes while increasing the margin to make the system more general and reliable. They work by putting raw data into high-dimensional spaces and finding the hyperplane that best divides the different groups. Because they use kernel functions, SVMs can be used for both linear and nonlinear decision limits. Their ability to work with high-dimensional data makes sure that classifications are correct even when there are a lot of features. This improves model performance and makes it more resistant to noise.

$$minimize \frac{1}{2} ||W||^2 + C \sum_{i=1}^n \xi_i \quad (4)$$

“Random Forest: RF is an ensemble ML method that takes the best parts of several DT” and mixes them together to make predictions more accurate and less variable. It makes it more resistant to noise and overfitting by building trees with random groups of features and data samples. The process of making decisions as a group makes sure that predictions are accurate and that generalization works well across different datasets. RF also show how important features are, which helps us understand what factors

are most important. This ensemble method makes the model more stable overall and always makes classification or regression work better.

$$Gini = 1 - \sum_{i=1}^c (P_i)^2 \quad (5)$$

Bagging Classifier: The Bagging Classifier uses bootstrap aggregating as part of an ensemble approach to make multiple models from training data subsets that were chosen at random. By putting these models' forecasts together, it lowers variance, stops overfitting, and makes the system more robust, especially for base learners that aren't stable, like decision trees. By using different model views, bagging makes the system more reliable and accurate at making predictions. This keeps performance stable even when the data changes. The way it works improves generalization while keeping things simple in ensemble learning apps.

Voting Classifier: Voting Classifiers take predictions from several base models and make a final choice based on consensus. This makes the system more accurate and reliable as a whole. This method can be used in both hard voting and soft voting. Hard voting selects the majority class, while soft voting bases the final prediction on the average of the class probabilities. Voting ensures that the system is more stable and general by exploiting the interplay of the strengths of each model by a number of different techniques. The ensemble mechanism is good in reducing bias and variance that contributes to making accurate predictions that are consistent over a large set of challenging classification problems.

$$\hat{y} = \underset{c}{\operatorname{argmax}} \left(\sum_{i=1}^n I(\hat{y}_i = c) \right) \quad (6)$$

4. EXPERIMENTAL RESULTS

Accuracy: The degree to which a test can distinguish between sick and healthy people is called the test's accuracy. To know the accuracy of a test, you need to look at the proportion of true-positive and true-negative cases. Mathematically it can be written as:

$$"Accuracy = \frac{TP + TN}{TP + FP + TN + FN} (7)"$$

Precision: Accuracy is the proportion of cases or samples that are correctly classified as positive compared to cases or samples that are correctly classified as positive. So, here's how to find the precision:

$$"Precision = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} (8)"$$

Recall: In ML, recall is a measure of how well a model can find all important instances of a given class. This shows how well the model captures instances of a particular class. It is calculated by dividing the number of correctly predicted positive observations by the total number of actual positive observations.

$$"Recall = \frac{TP}{TP + FN} (9)"$$

F1-Score: F1 score is a way to evaluate the accuracy of an ML model. Get the model's precision and recall scores and add them together. Accuracy metrics count the number of times a model makes a correct estimate across a dataset.

$$F1 \text{ Score} = 2 * \frac{Recall \times Precision}{Recall + Precision} * 100 (10)$$

AUC-ROC Curve: The AUC-ROC curve shows how well a classification problem is solved at different benchmark levels. The true positive rate is plotted by ROC against the false positive rate. AUC

measures how well a model can distinguish between classes. A higher AUC means better performance of the model.

$$AUC = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i) \cdot \frac{TPR_{i+1} + TPR_i}{2} \quad (11)$$

MAE: Absolute Error is the amount of mistake when you measure something. It's the difference between what was recorded and what was "true." To give you an example, if the scale says you weigh 90 pounds but you know you really weigh 89 pounds, the scale is off by 1 pound.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (12)$$

“Table.1 Performance Evaluation Table”

ML Model	Accuracy	f1_score	Recall	Precision	AUC	MAE
KNN	0.483	0.476	0.483	0.538	0.969	1.483
Naive Bayes	0.534	0.537	0.534	0.591	0.892	1.278
DecisionTree	0.926	0.926	0.926	0.932	1.000	0.080
SVM	0.068	0.128	0.068	1.000	0.000	2.324
Random Forest	0.972	0.972	0.972	0.975	1.000	0.051
Bagging Classifier	0.977	0.977	0.977	0.979	1.000	0.045
Voting Classifier	1.000	1.000	1.000	1.000	1.000	0.000

In Table.1, the performance of different ML models is “shown by Accuracy, F1-Score, Recall, Precision, AUC, and MAE. The Voting Classifier does better than all of them”, with perfect scores in every single measure.

Fig.3 Comparison Graph

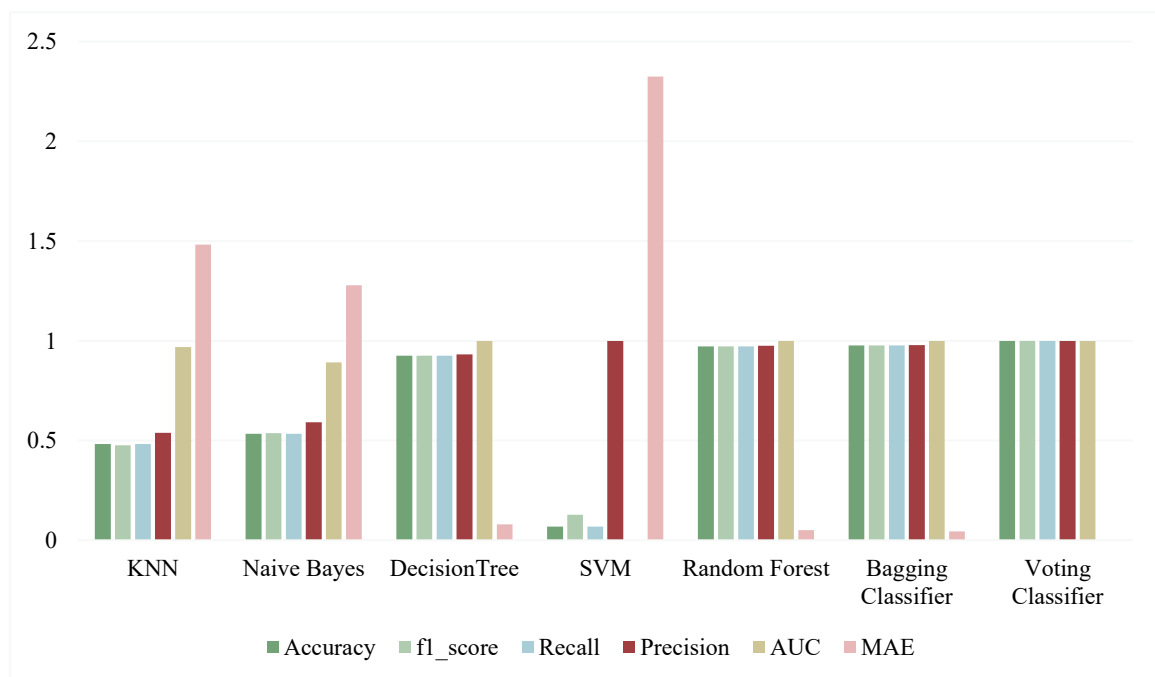


Figure 3 shows how different ML models did in “terms of Accuracy, F1-Score, Recall, Precision, AUC, and MAE. It shows that the Voting Classifier” did the best, with perfect scores for all of these metrics.

5. CONCLUSION

The system shows how ML can be used to accurately guess which crops will do well by looking at large sets of data about the soil and its environment. The pH, texture, nutrients, temperature, rainfall, and amount of nitrogen and phosphorus in the fertilizer are all key parts of these sets. The MRFE technique helps in identifying and prioritizing the most relevant parts of the system in a short period of time. This minimizes the variables of less importance and increases the prediction capability of the models. By applying a set of different classifiers to the features that were chosen, we have found out the commodities that would be successful in each area. Some of the algorithms included “K-Nearest Neighbors (KNN), Naive Bayes, Decision Tree, Support Vector Machine (SVM), Random Forest, Bagging Classifier, and Voting Classifier. To measure performance, we used such metrics as Accuracy (ACC), Precision, Recall, Specificity, F1-Score, Area Under the Curve (AUC), Mean Absolute Error (MAE)”, and Log Loss. The findings showed that the model was more consistently and effectively worked when features were selected by MRFE. The Voting Classifier was the most successful one with the highest success rate of 100 percent which implies that the Voting Classifier was stable, generalized, and accurate with its predictions. By combining MRFE and ML, a specific, fast, and scalable way to determine the success of certain commodities in a particular region is obtained. This is useful in making informed decisions about agriculture based on data.

In future, the technique can be improved through the use of bigger and more diverse datasets that offer more variation of the soil properties, climatic

conditions, and types of crops. This would make forecasts more accurate and practical in greater number of scenarios. One can improve the performance of the model by adding more such complex ensemble and DL approaches as Gradient Boosting, XGBoost, or LSTM networks. The IoT sensors that track the weather and soil conditions in real time can be used to help in determining the best produce to produce in a given area. Along with that, the system can be extended to give great advice on the matters of crop rotation, watering, and fertilization that can help form the sustainable and accurate agricultural practices.

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